



Foundations of Programming

Assignment No.: 22

PROBLEM STATEMENT: Write a menu driven program in C to perform various string operations using in-built functions.

OBJECTIVE:

- To understand the concept of **strings in the C programming language**.
- To learn how to use **built-in string functions** for string manipulation.
- To perform operations such as **length calculation, copying, concatenation, comparison, and reversal** using menu options.

THEORY:

In the C programming language, a **string** is a sequence of characters terminated by a null character ('\0'). Strings are stored in **character arrays** and manipulated using functions available in the **string.h header file**.

Common **built-in string functions** include `strlen()` for finding the length of a string, `strcpy()` for copying one string to another, `strcat()` for concatenating two strings, and `strcmp()` for comparing two strings.

A **menu driven program** allows the user to select different operations from a list of options. Based on the user's choice, the corresponding string function is executed. This approach makes the program interactive and easy to use.

PLATFORM: Linux

CODE:

```
#include<stdio.h>
```

```
#include<string.h>
```

```
int main()
```

```
{  
  
    char str1[100],str2[100],rev[100];  
  
    int ch;  
  
  
    printf("Enter first string: ");  
  
    scanf("%s",str1);  
  
  
    do{  
  
        printf("\n1.Length\n2.Reverse\n3.Compare\n4.Palindrome\n5.Substring\n6.Exit\n");  
  
        printf("Enter choice: ");  
  
        scanf("%d",&ch);  
  
  
        switch(ch)  
  
        {  
  
            case 1:  
  
                printf("Length = %lu\n",strlen(str1));  
  
                break;  
  
  
            case 2:  
  
                strcpy(rev,str1);  
  
                strrev(rev);  
  
                printf("Reverse = %s\n",rev);  
  
                break;
```

case 3:

```
printf("Enter second string: ");  
scanf("%s",str2);  
if(strcmp(str1,str2)==0)  
    printf("Strings Equal\n");  
else  
    printf("Strings Not Equal\n");  
break;
```

case 4:

```
strcpy(rev,str1);  
strrev(rev);  
if(strcmp(str1,rev)==0)  
    printf("Palindrome\n");  
else  
    printf("Not Palindrome\n");  
break;
```

case 5:

```
printf("Enter substring: ");  
scanf("%s",str2);  
if(strstr(str1,str2)!=NULL)  
    printf("Substring Found\n");  
else
```

```
        printf("Substring Not Found\n");  
        break;  
    }  
    }while(ch!=6);  
}
```

INPUT:

```
Enter first string: madam  
Choice: 1 2 4 6
```

OUTPUT:

```
Length = 5  
Reverse = madam  
Palindrome
```

SAMPLE INPUT:

```
Enter first string: Hello  
Enter second string: World  
Choice: 3
```

SAMPLE OUTPUT:

Concatenated string: HelloWorld

CONCLUSION: Thus, the C program successfully demonstrates **different string operations using built-in functions** through a **menu driven approach**, making the program interactive and easy to use.

FAQs:

1. What are **built-in string functions** in C and why are they used?
2. What is the purpose of the **string.h header file** in C programs?
3. How does a **menu driven program improve user interaction** in programming?
- 4.